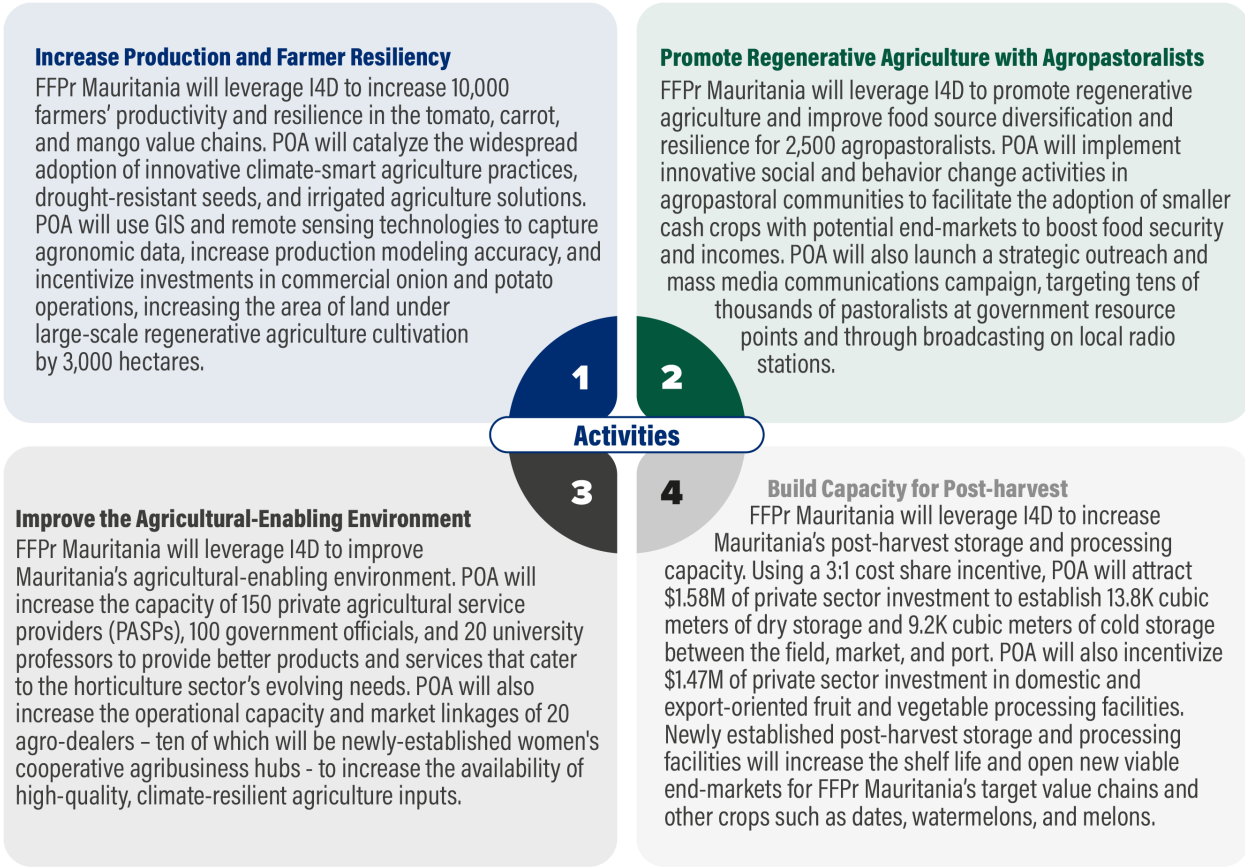
PLAN OF OPERATIONS

Food for Progress Mauritania (FFPr Mauritania) will provide a sustainable, reliable source of fruits and vegetables to help meet Mauritania's food security needs while increasing the resiliency of farmers to evolving climate conditions. FFPr Mauritania, implemented by Partners of the Americas (POA), will also expand US-Mauritania trade, increasing US exports to Mauritania by $14.8-17.3M over the five-year program. POA will implement FFPr Mauritania by employing a cost-effective strategy that leverages innovations, incentives, and private sector investments to modernize and optimize Mauritania's horticulture sector. We call this strategy **I4D** – ***Innovations, Incentives, and Investments for Development*** – and we leverage it throughout FFPr Mauritania's activities (Figure 1).

Fig 1: FFPr Mauritania's Activities Leverage I4D



FFPr Mauritania's target beneficiaries are primarily along the Senegal River Valley in Trarza, Brakna, Gorgol, Assaba, and Guidimakha. From inception to closeout, FFPr Mauritania will be a vehicle to advance gender and racial justice and equality. Activities target 21 high-functioning women's cooperatives and 8,200 farmers hailing from groups suffering systemic discrimination throughout Mauritania.

FFPr Mauritania will execute a cost-effective technical approach that capitalizes on Mauritania's advantageous natural resources and untapped private sector investors to increase agricultural productivity and expand the trade of agricultural products sustainably. FFPr Mauritania leverages Collaborating, Learning, and Adapting principles and builds on the success of agricultural projects implemented by the Government of Mauritania, the International Fund for Agricultural Development, the World Bank, USAID, WFP, and the African Development Bank. POA co-designed activities with key stakeholders and project partners to be flexible,

adaptable, and maximize lasting impact. Technical rigor and impactful partnerships will ensure that all activities are technically and politically feasible in the life of project (Tables 1 & 2).

Table 1: FFPr Mauritania Timeline of Anticipated Activity Start and Completion Dates

Activity	Sub Activity	FY24				FY25				FY26				FY27				FY28			
		Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
Start-up		■	■																		
Closeout																				■	■
Activity 1 *Increase production & Farmer Resiliency*	*1.1*			■	■	■	■	■	■	■	■										
	1.2	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■
	1.3			■	■	■	■	■	■												
	1.4				■	■	■	■	■												
Activity 2 *Promote Regenerative Agriculture with Agropastoralists*	*2.1*			■	■	■	■	■	■	■	■	■	■	■	■	■	■				
	2.2				■	■	■	■													
	2.3			■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■		
	2.4						■	■	■	■	■	■	■	■	■	■	■				
Activity 3 *Improve the Agricultural-Enabling Environment*	*3.1*					■	■	■	■	■	■	■	■								
	3.2			■	■	■	■	■	■	■	■	■	■	■	■	■	■				
	3.3			■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■		
	3.4					■	■	■	■	■	■	■	■	■	■	■	■	■			
	3.5			■	■	■	■	■	■	■	■	■	■	■	■	■	■	■	■		
	3.6					■	■	■	■	■	■	■	■	■	■						
	3.7					■	■	■	■	■	■	■	■	■	■						
Activity 4 *Build Capacity for Post-harvest*	*4.1*				■	■	■	■	■	■	■	■	■	■	■	■	■	■	■		
	4.2							■	■	■	■	■	■	■	■	■	■				
	4.3				■	■	■	■	■	■	■	■	■	■	■						
	4.4									■	■	■	■	■	■	■	■	■	■		
	4.5							■	■	■	■	■	■	■							

Table 2: Key Activity Partners

Partner	Description and FFPr Mauritania Role
6th Grain	Innovative GIS and remote sensing company with a presence in Mauritania that will build data usage capacity and boost farmer productivity.
African Agriculture	Sustainable rural development company working with 18 agriculture cooperatives and the Government of Mauritania to develop 2,000 hectares near Bogué, where FFPr will model improved horticultural production and forage.
Global Cold Chain Alliance	Leading global cold chain experts that will conduct a green-energy cold chain feasibility study to facilitate investment in Mauritanian agribusinesses and enable post-harvest capacity through cold chain storage improvements.
Higher Education Institute of Technology (ISET)	School offering degrees and training in agricultural science that will co-develop and deliver private agricultural service provider (PASP) training and whose students will participate in FFPr internship and capacity building programs.
Kaedi Agriculture Schools	Two agriculture schools serving primarily female students that will participate in the PASP, internship, and capacity building programs and have job opportunities with African Agriculture in commercial horticulture and forage production.
Lycée Agricole	Agriculture school offering agriculture and horticulture degrees to students ages 18-60 that will participate in the PASP, internships, and capacity building program.
National Horticulture Trade Association (UNIH)	Association of 17,000 Mauritanian horticulture farmers and cooperatives that will help identify "early adopter" horticulture farmers to train, take on key FFPr activities for program sustainability, and disseminate information and best practices produced during the project.

I. Activity 1 – Increase Production and Farmer Resiliency

1.1: Increase Farmer Productivity through Climate-Smart Agricultural Practices

- **Activity Lead:** POA
- **Target regions**: Trarza, Brakna, Gorgol, Assaba, and Guidimakha
- **Beneficiaries:** Farmers

As discussed in the Strategic Analysis, FFPr Mauritania will develop the tomato, carrot, mango, potato, and onion value chains – crops with viable domestic and/or export end-markets. Mauritanian farmers grow these crops but use practices that result in low yields and leave them vulnerable to climate change. FFPr Mauritania will disseminate climate-smart agriculture practices to key farmers in the Senegal River Valley beginning in Year 1. Activity 1.1 will result in Increased Knowledge by Farmers of Improved Agricultural Techniques and Technologies (FFPr 1.2.4) and Increased Access to Improved Market Information (FFPr 1.4.3; FFPr 2.4.3).

1.1.1: Develop Technical Package

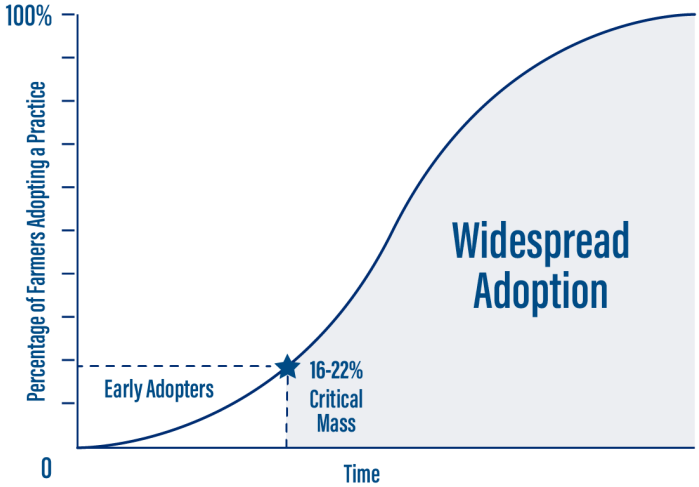
In Q3 of Year 1, POA will contract top-level agronomists and horticulturalists to develop technical packages that leverage the best climate-smart and regenerative agricultural practices for target crops. Topics in the extension package will include using drought-resistant seeds and cultivars, sustainable soil fertility management, intercropping and agroforestry, integrated pest management, precision fertilizer usage, on-farm experimentation, and digital technologies.

1.1.2: Identify "Early Adopter" Farmers

Mauritania has approximately 10,000 horticulture farmers in the Senegal River Valley. Most farmers have low production and are vulnerable to climate change. Training all farmers in climate-smart agriculture practices to increase farmer resiliency is not cost-effective. However, research indicates that once a critical mass of the target population adopts a new practice (16-22%), the practice will spread naturally through a chain reaction (Rogers, 2003). Therefore, in Q4 of Year 1, POA will identify 2,000 farmers that exhibit the characteristics of "early adopters" to reach critical mass more efficiently in the target regions (Figure 2). "Early adopters" typically have high social status and access to finance and are often young, educated, and willing to take moderate risks (Rogers, 2003). This will increase cost-effectiveness.

Focusing training on early adopters will maximize the cost-effectiveness of USDA's investment, accelerate the widespread adoption of climate-smart agriculture practices, and increase farmer resiliency throughout the sector.

Fig 2: Reaching a Critical Mass for Widespread Adoption



1.1.3: Conduct Agriculture Extension

1) Implement Traditional Extension

FFPr Mauritania will use the Farmer Field School approach to deliver technical packages to enhance farmer education. The project will collaborate with "early adopter" farmers to demonstrate the technical package on

their farms. To facilitate the process, FFPr Mauritania will cost share inputs and provide technical assistance to prepare the farms for the training sessions. POA will also distribute digital and printed versions of the technical package to Farmer Field School attendees, the Ministry of Agriculture, the National Horticulture Trade Association, and other relevant stakeholders. The FFPr Mauritania project-funded extension agents will implement the traditional extension activities from Q4 of Year 1 to Q2 of Year 3, after which the extension agents will transition to PASPs.

UNIH: A Critical Partner

Mauritania's National Horticulture Trade Association (UNIH) is comprised of 17,000 farmers and 3,000 farmers groups, associations, and cooperatives. They signed an exclusive partner agreement with POA and helped design FFPr Mauritania. POA will leverage the National Horticulture Trade Association's knowledge of the agricultural landscape in Mauritania to accelerate start-up and quickly identify "early adopter" beneficiaries.

2) Transfer Extension Responsibilities to PASPs

By Year 3, project extension agents and other certified PASPs (Activity 3.2) will have the necessary skills to provide fee-for-service extension beyond the life of the project. To achieve this, FFPr Mauritania extension agents will complete FFPr Mauritania's PASP Certification Program in Year 2 (Activity 3.2). These certified PASPs will provide fee-based extension services and inputs such as high-quality drought-resistant hybrid seeds and fertilizer.

3) Promote Digital Agriculture Extension

To reach younger farmers and ensure support when traditional extension services and PASPs are unavailable, POA will integrate digital extension methods into capacity building activities throughout the project. Digital extension services will include information such as market prices, demand, weather reports, pest and disease spread, water supply, and input availability. Promoting digital services will be modeled after programs in neighboring Senegal, such as AICCRA, Winrock International's Nafoore Warsaaji Project, Mlouma, and Jokalant. Digital extension services can foster mutually beneficial relationships by creating linkages between PASPs and farmers. Farmers receive inputs and technical assistance, while PASPs gain more customers and business opportunities.

1.2: Incentivize Large-Scale Onion and Potato Production

- **Activity Lead:** POA and 6th Grain
- **Target Regions:** Trarza and Brakna
- **Beneficiaries:** Commercial Farmers

Large-Scale Agriculture is the Key

Many agricultural development projects in developing countries focus exclusively on the small-scale farmer due to topography and land distribution, among other reasons. USDA has a unique opportunity to also invest in Mauritania's untapped potential for large-scale, commercial agriculture. While increasing small-scale farmer productivity is important, large-scale commercial agriculture is crucial to significantly decreasing reliance on imports and increasing food security in Mauritania.

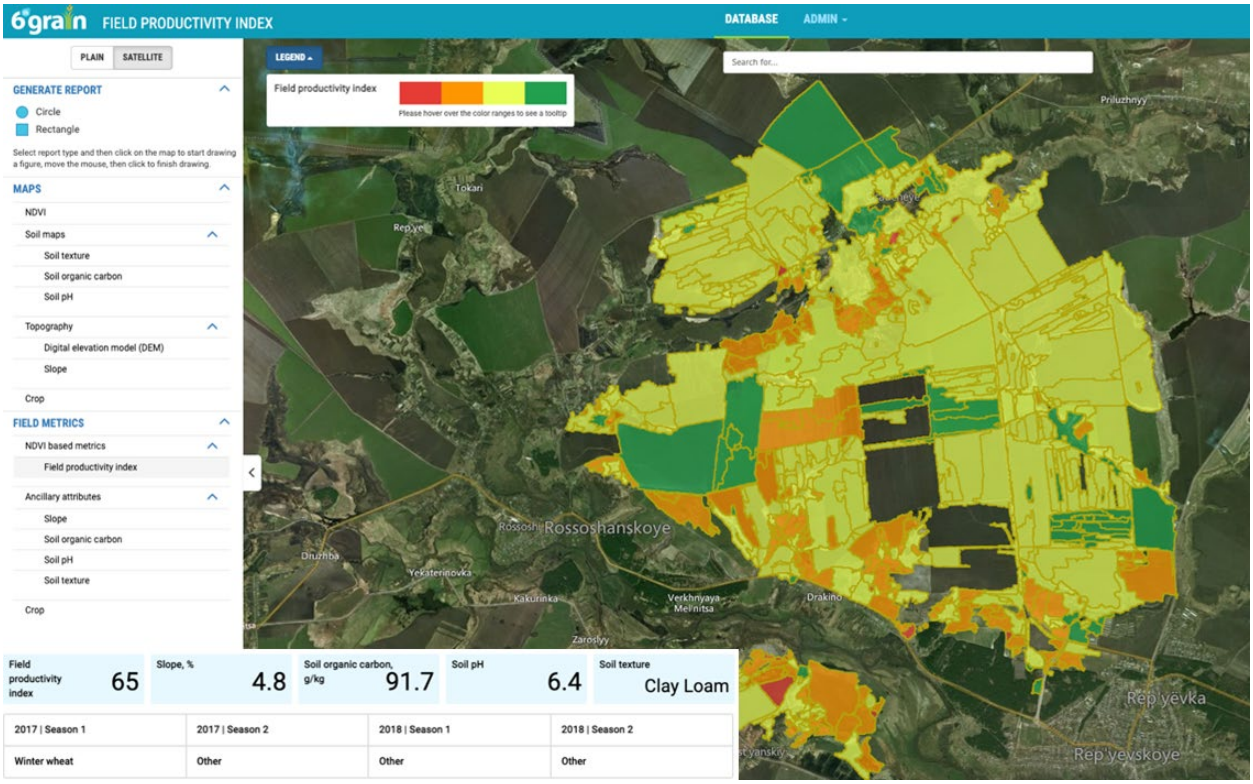
As mentioned in the Strategic Analysis, the agriculture sector in Mauritania must drastically increase the production of staple foods such as potatoes and onions to address chronic food insecurity. POA's field research revealed significant interest from Nouakchott-based investors to open large-scale farms in Mauritania. However, many of these potential investors remain wary due to risks associated with production potential, profitability, and land tenure. Increasing the availability of production and climate data will reduce uncertainties and risks for large-scale potato and onion farming in Mauritania. Partnering with 6th Grain, a US-based GIS and remote sensing company, POA will increase critical data availability to incentivize investment in large-scale commercial agriculture. Activity 1.2 will lead to Improved

Knowledge Regarding Farm Management (FFPr 1.3.1), Increased Access to Improved Market Information (FFPr 1.4.3; FFPr 2.4.3), and Increased Leverage of Private-Sector Resources (FFPr 1.4.5; FFPr 2.4.5).

As part of FFPr Mauritania's baseline study, in Q1 of Year 1, POA and 6th Grain will create a digital registry of Mauritania's horticulture farmers and use GIS and remote sensing tools to create a dynamic map of the country's horticulture production. This tool will be widely available, allowing government agencies, PASPs, NGOs, and others to identify opportunities for high-impact interventions in real time. For example, the map will function as an early warning sign for outbreaks of pests and diseases, allowing for rapid responses and effective interventions.

6th Grain will also use this data and meteorological data to model crop production potential for large-scale onion and potato production. They will build integrated models that monitor and predict crop growth, nutrient requirements, and pests and disease risks with at least 85% accuracy (Figure 3). Starting in Q2 of Year 1, POA will present the data to Nouakchott-based investors, pitching them the low-risk, high-reward agribusiness opportunities in large-scale, commercial onion and potato farming.

Fig. 3: Illustrative Field Productivity Index Dashboard



FFPr Mauritania's Field Productivity Index Dashboard will generate real-time agronomic data from Mauritania's horticulture sector. POA will present this to investors to incentivize large-scale, commercial onion and potato production. The data will also be an asset for PASPs to use to identify needs and potential clients in the horticulture sector. Furthermore, POA will present the data at investment roundtables (Activity 4.1) to illustrate production levels that a post-harvest processing facility could anticipate. This dashboard is sustainable due to profit incentives. Multinational agribusinesses, such as Corteva and Syngenta, will use this real-time data to manage seed, fertilizer, and crop protection inventories in Mauritania. They currently pay 6th Grain to maintain such donor-funded dashboards in Uganda and Egypt. Therefore, once FFPr Mauritania closes, 6th Grain will continue to maintain this public-facing dashboard sustainably.

1.3: Introduce New and Innovative Irrigation Solutions

- **Activity Lead:** POA
- **Target Regions:** Trarza, Brakna, and Gorgol
- **Beneficiaries:** Farmers and Agro-Dealers

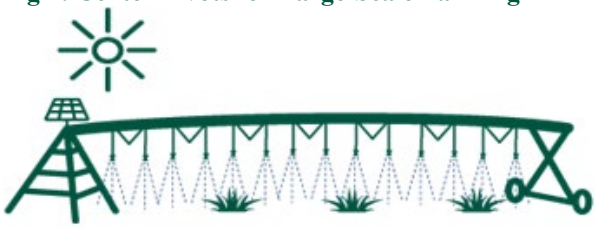
As discussed in the Strategic Analysis, enhancing irrigation methods and practices is crucial for Mauritanian farmers to sustainably increase the productivity of tomato, carrot, mango, potato, and onion cropping systems. POA will coordinate with other irrigation-focused programs, including World Bank PARIIS and International Fund for Agricultural Development's PRODEFI and PROGRES, to showcase innovative irrigation solutions. Activity 1.3 will lead to Increased Knowledge by Farmers of Improved Agricultural Techniques and Technologies (FFPr 1.2.4) and Improved Knowledge Regarding Farm Management (FFPr 1.3.1).

Starting in Q3 of Year 1, POA will work with high-capacity farmers and agricultural institutions to demonstrate innovative irrigation techniques such as drip and sprinkler aspersion. POA will integrate green energy options, such as solar and wind power, into all demonstrations. Farmers will learn the benefits of gravity-fed irrigation, which uses green energy and minimizes dependence on diesel.

FFPr Mauritania will leverage the Farmer Field School approach to target small-holder farmers and cooperatives at two sites in the Trarza, Brakna, and Gorgol regions (six total). The demonstration plots and Farmer Field School days will run from Q4 of Year 1 to Q4 of Year 2. The plots will serve to educate farmers on the advantages and disadvantages of each technique depending on their specific plot, local environment, and economic means. Best practices and examples from other successful development programs, such as World Bank PARIIS and IFAD's PRODEFI and PROGRES, will be used to design and implement demonstration plots.

To enhance large-scale commercial farming in Mauritania, FFPr will demonstrate the center pivot irrigation method (Figure 4). POA will collaborate with African Agriculture starting in Q4 of Year 1 to showcase the efficacy of center pivots for large-scale onion and potato cultivation.

Fig 4: Center Pivots for Large-Scale Farming



The center pivot is a mobile sprinkler system that is easy to install and creates a heat-resistant microclimate.

1.4: Bolster Existing Underutilized Water Infrastructure

- **Activity Lead:** POA
- **Target Regions:** Trarza, Brakna, Gorgol, Assaba, and Guidimakha
- **Beneficiaries:** Farmers, Ministry of Agriculture, and Agribusinesses

As described in the Strategic Analysis, many irrigation infrastructure projects in Mauritania are not operating at full capacity. To enhance their effectiveness, POA has developed a strategy to troubleshoot these projects and identify innovative methods to enhance their design, mechanics, and management. Rather than simply upgrading underutilized water infrastructure, which would not be a cost-effective use of USDA funds, FFPr Mauritania will target systemic improvements for irrigated agriculture operations. POA will accomplish this through collaboration with key partners such as the Ministry of Agriculture, the World Bank, and the International Fund for Agriculture Development. Activity 1.4 will lead to Improved Infrastructure to Support On-Farm Production (FFPr 1.2.2), Increased Knowledge by Farmers

of Improved Agricultural Techniques and Technologies (FFPr 1.2.4), and Improved Knowledge Regarding Farm Management (FFPr 1.3.1).

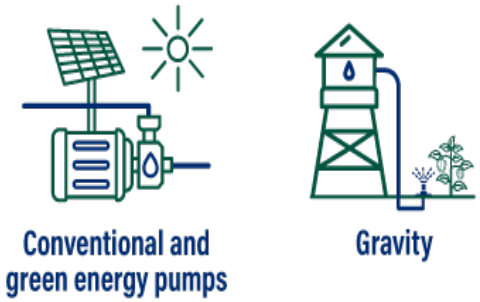
1.4.1: Improve On-farm Irrigation Capacity

In Q4 of Year 1, POA staff and a top-tier irrigation consultant will visit farms to evaluate infrastructure installed by current and previous development projects to improve on-farm irrigation. This process will involve partnering with irrigation initiatives such as the World Bank PARIIS, which specializes in adapting horticulture pumps for the Mauritanian environment. POA will generate a practical report on its discoveries to disseminate to farmers, the government, and other development programs in Q2 of Year 2. Additionally, POA will instruct PASPs on practical techniques for setting up, reengineering, and refurbishing on-farm irrigation equipment, such as pumps and tanks (Figure 5).

1.4.2: Upgrade Large-Scale Water Infrastructure

During Q2 of Year 2, FFPr Mauritania will collaborate with other donor-funded programs and relevant government agencies, including the Ministry of Agriculture and SONADER, to address issues associated with existing underperforming irrigation projects. For example, SONADER, the government's water infrastructure agency, struggles to adequately manage significant large-scale irrigation perimeters. Irrigation perimeters are the source for most on-farm horticulture irrigation and must be reliable and effectively managed to unlock productivity. Expert technical irrigation hydrology engineers and project management staff will troubleshoot problems such as clogged canals, deteriorating equipment, insufficient water flow, poor maintenance, inadequate community management systems, and ineffective fee systems. FFPr Mauritania will publish and disseminate innovative solutions and findings in Q4 of Year 2.

Fig 5: Green-Energy Irrigation



FFPr Mauritania will promote green energy solutions for on-farm irrigation such as reconfiguring conventional pumps to use solar and wind power. POA will also integrate gravity irrigation that reduces energy needs and provides a reliable source of water.

II. Activity 2 – Promote Regenerative Agriculture with Agropastoralists

2.1: Increase Small Cash and Tree Cropping in Agropastoral Communities

- **Activity Lead:** POA
- **Target Regions:** Trarza, Brakna, and Gorgol
- **Beneficiaries:** Agropastoralists

As the Strategic Analysis mentions, horticulture production provides agropastoral communities with opportunities to diversify their food and livelihood sources. Crops such as mangoes, tomatoes, and carrots have strong domestic and regional demand and can improve household nutrition. Additionally, regenerative agriculture practices mitigate desertification, build long-term soil health, and improve the yields of small cash crops. Activity 2.1 will lead to Increased Knowledge by Farmers of Improved Agricultural Techniques and Technologies (FFPr 1.2.4), Improved Knowledge Regarding Farm Management (FFPr 1.3.1), Improved Marketing of Agricultural Products (FFPr 2.1.3;2.2.1), and Increased Access to Improved Market Information (FFPr 1.4.3; FFPr 2.4.3).

2.1.1: Boost Horticulture Cropping Systems

In Q3 of Year 1, FFPr Mauritania will begin offering training on small-scale regenerative agriculture opportunities to agropastoral families from 20 communities. Target beneficiaries will be households with permanent homes near the Senegal River. Training will focus on smaller cash crops, forage, and tree crops with end-markets in regional hubs such as Rosso, Kaedi, Bogue, and Nouakchott. POA will promote agroforestry by combining short-cycle vegetable crops with tree crops with viable end-markets, such as mango, lime, and papaya. Capacity building will include market identification and linkages for crop sales. This will increase household food security and diversify income sources for improved resilience. The training sessions will run until Q4 of Year 4 and target agropastoralists with little to no experience in horticulture production. POA has identified relevant demonstration plots and will use sites created in Activity 1.1. POA will conduct training activities in collaboration with the Ministries of Agriculture and Livestock and the World Bank PRAPS II agropastoralist project.

World Bank PRAPS Phase II

POA plans to implement activities with the World Bank PRAPS project that is increasing the resilience of agropastoral communities in Mauritania. PRAPS and POA will collaborate to promote horticulture with women and youth in agropastoral communities along the Senegal River. World Bank staff will share lessons learned on food source and livelihood diversification and provide guidance on working with cooperatives in Fulani minority communities.

2.1.2: Facilitate Commercial Forage Production

FFPr Mauritania will promote commercial forage production that reduces desertification. In Q1 of Year 2, POA will partner with African Agriculture, a US-based agribusiness growing alfalfa on 2,000 hectares of land near Bogue. POA and African Agriculture will work with agropastoralists to recruit them in outgrower models that produce high-quality forage.

2.2: Reduce Desertification with Horticultural Tree Buffer Zones

- **Activity Lead:** POA
- **Target Regions:** Trarza, Brakna, and Gorgol
- **Beneficiaries:** Agropastoralists and Farmers

As discussed in the Strategic Analysis, desertification is a persistent threat to Mauritanian food security and sustainable development. POA will use horticultural tree buffer zones to stop wind and soil erosion and improve soil quality. Additionally, farmers will increase yields and diversify livelihoods by incorporating smaller cash crops, such as tomatoes, and tree crops with forage in the horticulture tree buffer zones. Activity 2.2 will lead to Increased Knowledge by Farmers of Improved Agricultural Techniques and Technologies (FFPr 1.2.4) and Improved Knowledge Regarding Farm Management (FFPr 1.3.1).

In Q4 of Year 1, POA will contract nurseries in Mauritania and Senegal to grow a total of 50,000 saplings of mango, lime, native acacia trees, and other species. In Q3 of Year 2, during the rainy season, POA will collaborate with 500 agropastoralists and farmers in Trarza, Brakna, Gorgol, Assaba, and Guidimakha to establish horticultural tree buffer zones (Figure 6). These zones will incorporate a mix of mango, lime, papaya, acacia, Napier grass, and other native trees around agriculture fields. FFPr will assist farmers in constructing short-term fences to safeguard edible species eaten by animals such as goats. Horticultural tree buffer zones, in some cases, will be planted in phases, beginning with planting thorny acacia trees to fully enclose an area,

followed by tree crops, fodder, and smaller cash crops in the live-fence-protected area. During Q1 and Q2 of Year 2, POA will train farmers on establishing and maintaining horticultural tree buffer zones. The training sessions will emphasize correct tree planting and maintenance methods, including pruning.

Fig 6: Horticultural Tree Buffer Zones

Horticultural tree buffer zones prevent desertification resulting from wind and soil erosion. They consist of rows of fruit tree crops, native species, and grasses. They unlock sustainable horticulture production in areas where it would otherwise be unsuitable. Planting buffer zones is an effective strategy to mitigate the effects of climate change and increase food security and resilience.

2.3: Conduct Outreach and Awareness Campaigns

- **Activity Lead:** POA
- **Target Regions:** Countrywide
- **Beneficiaries:** Sedentary and Nomadic Pastoralists

As indicated in the Strategic Analysis, there are opportunities to promote horticulture activities within agropastoral communities. Diversifying food sources is critical for agropastoralists to rely less on livestock and improve their livelihood and nutrition. POA will implement strategic mass media outreach to communicate this effectively, including radio outreach. Radio is widely used by agropastoralists daily, with 80% having radio access. FFPr Mauritania will collaborate with the Ministry of Livestock and other branches of the Government of Mauritania to develop messaging promoting diverse food sources. POA will collaborate with PRAPS and build inter-ethnic collaboration over water, land, and transhumance corridors. Activity 2.3 will lead to Increased Knowledge by Farmers of Improved Agricultural Techniques and Technologies (FFPr 1.2.4)

POA will launch a radio and text messaging campaign to promote the advantages of diversifying food sources. In Q3 of Year 1, POA will design and develop the campaign with leaders and youth in agropastoral communities. The campaign will be broadcast from Q2 of Year 2 until Q2 of Year 5. To maximize lasting impact, the messaging will be broadcast in multiple languages, including Hassaniya, Pulaar, and Wolof. This campaign will cover topics such as nutrition, horticulture livelihood opportunities, land laws regarding livestock and agriculture in the Valley, and the impact of climate change on natural resources crucial to livestock production. To assess the effectiveness of the communications campaign, POA will employ Collaborating, Learning, Adapting, and other related principles.

2.4: Diversify Food Sources through Training and Social Behavior Change Activities

- **Activity Lead:** POA
- **Target Regions:** Trarza, Brakna, and Gorgol
- **Beneficiaries:** Agropastoralists

As described in the Strategic Analysis, more agropastoralists are permanently settling in the Senegal River Valley due to the available water resources and the loss of traditional grazing lands. These populations are increasingly vulnerable to climate change and food insecurity. To address these challenges, POA plans to conduct social behavior change activities to promote adopting sustainable agricultural practices with around 20 primarily minority Fulani communities. FFPr Mauritania will demonstrate economically feasible and culturally

appropriate horticulture activities. This will improve food security and build resilience to climate change. Activity 2.4 will lead to Increased Knowledge by Farmers of Improved Agricultural Techniques and Technologies (FFPr 1.2.4).

POA will build on its decades of successful work using social behavior change approaches with marginalized communities to catalyze Mauritanian agropastoralists to become more resilient and food secure in the target regions. To deliver effective nutrition education messaging, POA will use Collaborating, Learning, and Adapting tools, including the transtheoretical and early adoption models cited in Activity 1.1.2. This outreach and sensitization effort will consider lessons learned from other donor projects' work with agropastoralists. POA will establish local exchange platforms and community groups to train community leaders, cooperatives, and other stakeholders in pastoral communities on good nutrition practices and the benefits of food diversity. POA and partners will facilitate cooking demonstrations using local produce in pastoral communities and distribute educational materials on beneficial food and nutrition practices. Activities will begin in Q2 of Year 2 and continue through Year 4, using the Farmer Field School education model.

Social and Behavior Change in Haiti

POA is working to address micronutrient deficiencies in Haiti by promoting behavior change. With a sophisticated, multimodal communications campaign, POA has improved knowledge about the benefits of fortified foods and increased consumption of fortified products. Through its USAID RANFOSE program, POA conducts trainings, school and university visits, and media outreach via radio, TV, and newspaper. In response to ongoing political unrest and natural disasters, POA has adapted these strategies to be implemented remotely as needed. FFPr Mauritania's social behavior change activities are informed by POA's experience implementing successful communications campaigns under severe resource constraints.

III. Activity 3 – Improve the Agricultural Enabling Environment

3.1: Improve and Increase Private Agro-Dealers

- **Activity Lead:** POA
- **Target Regions:** Trarza, Brakna, and Gorgol
- **Beneficiaries:** Women's Cooperatives, Agro-Dealers, and Farmers

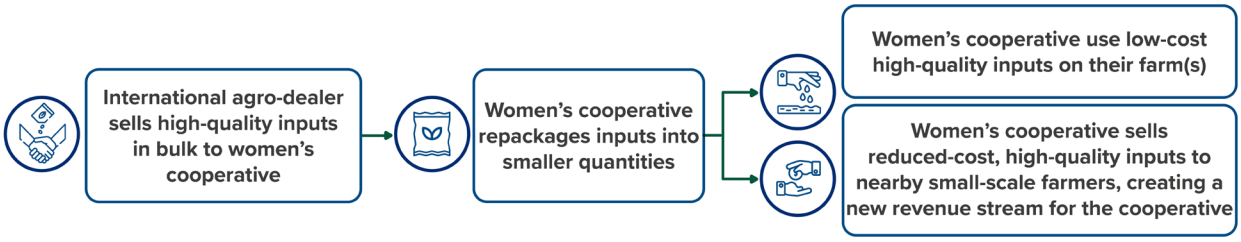
As mentioned in the Strategic Analysis, making high-quality and affordable agricultural inputs more available is essential to scale Mauritania's agriculture sector productivity. One way to achieve this is by improving the capacity of current agro-dealers and establishing women's cooperatives as "Agribusiness Hubs." FFPr Mauritania will leverage I4D to increase the availability of inputs by building agro-dealers' business capacity and creating market linkages between input suppliers and local enterprises. Activity 3.1 will lead to Increased Availability of Improved Inputs (FFPr 1.2.1), Improved Knowledge Regarding Farm Management (FFPr 1.3.1), and Improved Capacity of Key Groups in the Agriculture Sector (FFPr 1.4.4).

3.1.1: Increase Women's Cooperatives' Capacity to Function as Agro-Dealers

To empower women in rural areas and enhance the quality of agricultural inputs in Mauritania, POA will facilitate market linkages and build the capacity of ten high-functioning women's cooperatives to become agro-dealers (Figure 7). They will purchase and sell high-quality seeds, fertilizer, and other inputs from reputable suppliers like Green Seeds, IRZ Irrigation, and Corteva Agriscience. Capacity building activities will focus on cooperatives near Rosso and Bogue, ensuring easy access to services, customers, and end-markets.

In Q1 of Year 2, FFPr Mauritania will support women's cooperatives to improve their storage infrastructure and purchase critical inputs for business start-ups. They will receive training in inventory management, profit forecasting, record keeping, and input storage. With training, these new businesses can offer similar support services to their clients, enabling them to improve their farm business and increase sales. POA will create business plans with cooperatives, including purchasing inputs in bulk to take advantage of economies of scale. This activity advances racial justice, equity, and inclusion by providing economic opportunities to different ethnic groups, women, and youth historically lacking access to resources.

Fig 7: Women's Cooperatives will Thrive as Agro-Dealers



3.1.2: Boost Existing Agro-Dealers' Capacity

POA will enhance the capabilities of promising early-stage agro-dealers in Rosso, Bogue, and Kaedi. In Q1 of Year 2, POA plans to conduct training on agro-dealer operations, such as sourcing high-quality seeds, interpreting pesticide labels, and marketing. POA will train agro-dealers who operate as PASPs to use remote sensing and GIS data produced with 6th Grain under Activity 1.2. PASP-agro-dealers will learn to identify the spread of pests and diseases, such as desert locusts, and deliver farmers plant protection products to tackle issues (see Sustainable PASP Model text box). Additionally, agro-dealers will receive training in business operations, including recordkeeping, budgeting, and sales forecasting. The program will equip agro-dealers with the skills and knowledge necessary to provide similar training to small-scale clients under the PASP model, ultimately improving farmers' ability to purchase essential inputs.

3.2: Implement PASP Certification Program

- **Activity Lead:** POA and the Higher Education Institute of Technology (ISET)
- **Target Regions**: Trarza, Brakna, Gorgol, Assaba, and Guidimakha
- **Beneficiaries:** PASPs, Government Extension Agents, and Farmers

As the Strategic Analysis describes, FFPr Mauritania will implement a PASP certification program to increase the number of extension agents with specialized horticulture training. POA will partner with the Higher Education Institute of Technology and the Ministry of Agriculture to develop a PASP certification program that meets the needs of farmers. The program will target women, youth, and individuals representing minority groups to ensure equitable diffusion of knowledge. The PASP certification program will draw its candidates from Lycée Agricole and the two Kaedi schools that promote agricultural production and business. PASPs will learn about agriculture production, business, and marketing. Activity 3.2 will lead to Improved Knowledge

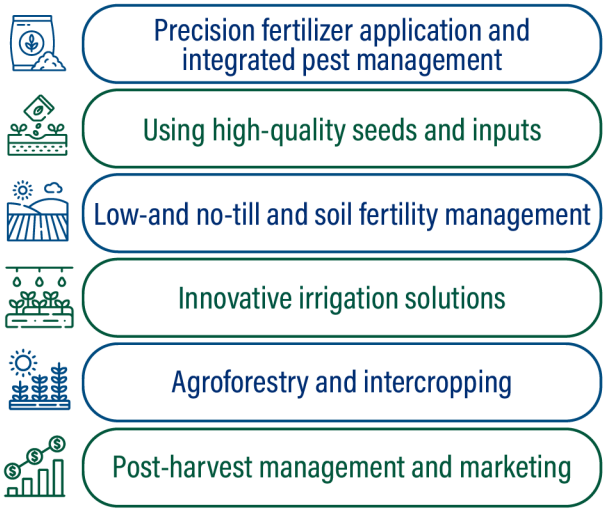
Sustainable PASP Model in Uganda

In Uganda, PASPs use 6th Grain remote sensing data to identify the spread of fungal disease outbreaks before it affects production. The PASPs bring the data to the farmers and provide them with technical assistance to resolve the issue. The data creates a win-win scenario for PASPs and farmers—the PASPs create business for themselves while the farmers mitigates losses. FFPr Mauritania will train PASPs how to use 6th Grain's technology to improve their business opportunities sustainably.

Regarding Farm Management (FFPr 1.3.1), Increased Capacity of Government Institutions (FFPr 1.4.1; 2.4.1), and Improved Capacity of Key Groups in the Agriculture Sector (FFPr 1.4.4).

In Q3 of Year 1, POA and the Higher Education Institute of Technology will develop a curriculum for a PASP certification program. POA will design modules focused on practical fieldwork, engaging local farmers and agribusinesses in design and implementation. The certification process will include six two-week climate-smart agriculture modules (Figure 8). The modules will focus on FFPr Mauritania target crops. Additional modules will cover business support services such as marketing and digital extension.

Fig 8: Illustrative PASP Learning Modules



There will be three PASP cohorts beginning in Q3 of Year 2. By the end of the project, 150 PASPs from the Valley will be certified – more than double the number of current extension agents. Certified PASPs will qualify for business start-up funding from the project. POA will also invite ministerial-level and local government extension agents to lead and participate in relevant modules. Specific activities will improve coordination between PASPs and public extension agents to ensure widespread coverage and sustain public and private investments in the agriculture sector.

3.3: Upgrade Agriculture Institutions' Technical Capacity

- **Activity Lead:** POA
- **Target Regions:** Trarza, Brakna, and Gorgol
- **Beneficiaries:** Higher Education Institute of Technology, Lycée Agricole, Kaedi Schools

As mentioned in the Strategic Analysis, domestic agriculture institutions must equip technicians of varying capacities with the foundational skills they need to meet the needs of farmers. To achieve this, POA will collaborate with key agriculture institutions to enhance the capabilities of their faculty and educational activities. The focus areas for improvement will include climate-smart agriculture production, mechanization, and agribusiness. POA will seek input from private sector partners, such as Mauritrac (CAT), John Deere, Corteva, and Green Seeds, to incorporate their business expertise into activities. Activity 3.3 will lead to Increased Leverage of Private-Sector Resources (FFPr 1.4.5; 2.4.5) and Improved Capacity of Key Groups in the Agriculture Sector (FFPr 1.4.4).

3.3.1: Improve Technical Institutions Capacity

POA will collaborate with African Agriculture and US colleges such as Michigan State University to build the technical capacity of Mauritanian agriculture institutions. POA and African Agriculture will collaborate to increase the capacity of two agriculture schools in Kaedi that primarily serve women. These women will have job opportunities to work with African Agriculture in commercial horticulture and forage production. FFPr Mauritania will share the technical package created under Activity 1.1 with schools to improve their climate-smart

agriculture technical capacity. POA will work with schools to integrate remote sensing and GIS data produced in Activity 1.2 into their curriculum and activities.

3.3.2: Build Mechanization Capacity

FFPr Mauritania will improve technical institutions' capacity to educate students in agricultural mechanization. The manager at Mauritrac (CAT) in Nouakchott told POA that "the most impactful agriculture sector mechanization intervention would be to increase the capacity of trainers" (Interview: Lamarque, 03/31/23). Starting in Q3 of Year 1, POA will facilitate faculty training at agriculture institutions in Mauritrac (CAT), John Deere, and Massey Ferguson machinery operations and maintenance. POA will tailor lessons to each school, company, and faculty member based on each program and product's needs. POA will drive market linkages and create opportunities to use US equipment at agriculture schools. These activities will increase US exports while improving the long-term economic sustainability of farms due to the high quality of these products (Figure 9).

Fig. 9: US Machinery is Critical

Director of Caterpillar's Nouakchott-based distributor, Ingrid Lamarque, told POA that 10% of their sales in Mauritania are heavy machinery used for agriculture. She is confident that FFPr Mauritania's inclusion of large-scale farmers will boost machinery rentals and sales. Mauritania will require a skilled workforce to operate this machinery.

3.4: Coordinate Agriculture Specialist Internships

- **Activity Lead:** POA
- **Target Regions:** Trarza, Brakna, and Gorgol
- **Beneficiaries:** Students, Commercial Farms, and Agribusinesses

As the Strategic Analysis mentions, an obstacle to attracting investment in new commercial, large-scale farms is a shortage of highly qualified agronomists and farm managers. Commercial farmers, investors, and agribusinesses repeatedly informed POA that graduates from domestic agriculture institutions lack the practical and specialized skills they seek in employees. POA will facilitate competitive internships with domestic and international horticulture sector actors. This will bridge the skills gap and provide students with practical training to meet employers' needs in the horticulture sector. Activity 3.4 will lead to Improved Capacity of Key Groups in the Agriculture Sector (FFPr 1.4.4) and Improved Knowledge Regarding Farm Management (FFPr 1.3.1).

POA will implement a competitive internship program for graduates of the Higher Education Institute of Technology, Lycée Agricole Bogue, and the Kaedi schools. The program will cover three cohorts of six interns (18 total), providing competitive six-month paid internships equivalent to entry-level Ministry of Agriculture extension jobs. Students will be empowered as highly specialized agronomists, machinery operators, and future agribusiness owners. Beginning in Q1 of Year 2, POA will place participants with large-scale domestic farms such as Rim Fruit, Sadarane Farms, and Africom SARL. Additionally, to access the wealth of horticulture knowledge available, POA will place some participants at international farms with similar agroecological zones and horticultural crops, such as Arizona, coastal Peru, and Egypt.

Similarly, POA will facilitate three-month international exchange programs for students to work and learn at elite international agriculture universities and companies with whom POA has strong relationships, such as UC Davis, Zamorano, John Deere, and Corteva. FFPr Mauritania will oversee the selection process, ensuring gender and ethnic balance among participants.

Selected students must commit to working in Mauritania for at least three years after completing the program to ensure that USDA's investment creates a lasting impact in Mauritania.

3.5: Facilitate Outgrower Model

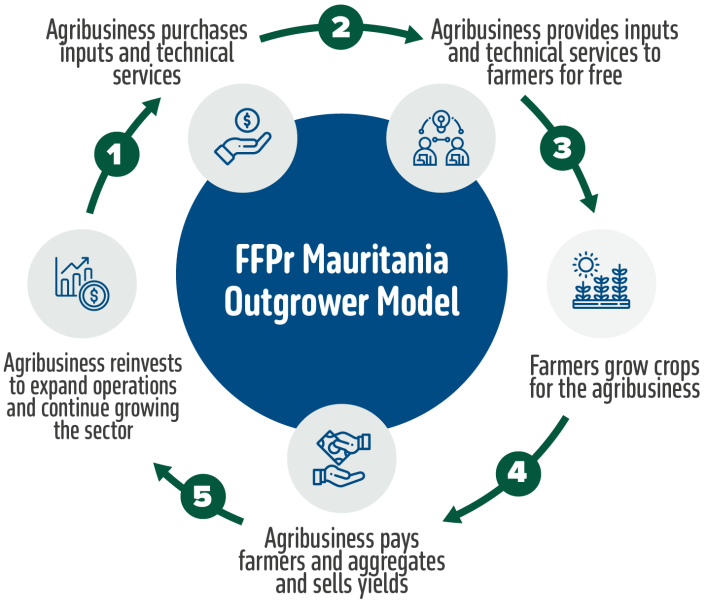
- **Target Regions:** Trarza and Brakna
- **Activity Lead:** POA
- **Beneficiaries:** Farmers, Agribusinesses, and the Government of Mauritania

As described in the Strategic Analysis, small-scale farmers in Mauritania face a significant challenge regarding limited access to inputs, finance, and technical assistance, which stunts their productivity and resilience. The outgrower model effectively overcomes these challenges by directing private sector resources to support small-scale farmers, aggregate their yields, and create economies of scale. POA will collaborate with the International Fund for Agricultural Development PRODEFI program, which successfully facilitates outgrower models between agribusinesses and small-scale farmers. Additionally, POA will enable government agencies and extension agents to continue facilitating outgrower models beyond the life of the project, thereby sustaining investment in the agriculture sector. Activity 3.5 will lead to Increased Availability of Improved Inputs (FFPr 1.2.1), Increased Use of Financial Services (FFPr 1.2.3; 2.2.3.1/2.3.1.1), Improved Marketing of Agricultural Products (FFPr 2.1.3; 2.2.1), Improved Linkages between Buyers and Sellers (FFPr 2.2.2), Increased Access to Improved Market Information (FFPr 1.4.3; FFPr 2.4.3), and Increased Leverage of Private Sector Resources (FFPr 1.4.5; 2.4.5).

Starting in Q3 of Year 1, POA will support outgrower models (Figure 10) for the tomato and carrot value chains. FFPr Mauritania will leverage private sector resources to provide approximately 500 farmers with inputs and finance for production. To achieve this, FFPr Mauritania will identify private sector partners interested in purchasing harvests from farmers and coordinate initial logistics. POA will then work with the private sector and the Ministry of Agriculture to organize and sensitize farmers on the benefits of the outgrower model and provide technical training on production. These activities will target minority groups historically excluded from development aid programs and youth lacking resources, thus promoting inclusive growth and development.

To ensure the success and sustainability of the outgrower model, POA plans to leverage best practices from the International Fund for Agricultural Development's PRODEFI project. These include using the Business-to-Business website https://beta.mr, which has proven to be an effective market linkage solution in Mauritania. Lessons learned on the outgrower model will be shared with the Ministry of Agriculture and other relevant government agencies to encourage embedding the outgrower model into their operations.

Fig. 10: FFPr Mauritania Outgrower Model



POA will facilitate connections between agribusinesses and farmers to catalyze productive partnerships. Agribusinesses will create economies of scale, while farmers will gain access to crucial inputs and technical assistance to increase productivity.

As traditional extension services are constrained in Mauritania, the Ministry of Agriculture can maximize impact by facilitating the outgrower model, serving as the liaison between the private sector and farmers. This will allow the government to maximize the resources at its disposal; the Ministry of Agriculture can promote the growth of the agriculture sector and provide small-scale farmers with access to finance, inputs, and markets.

3.6: Launch the Mauritanian Ag Expo

- **Activity Lead:** POA and National Horticulture Trade Association
- **Target Regions:** Rosso
- **Target Beneficiaries:** Farmers, Agribusiness, and the Government of Mauritania

As stated in the Strategic Analysis, the horticulture sector in Mauritania requires spaces to stimulate investment and introduce farmers and agribusinesses to new practices and products. Establishing an agricultural expo is a cost-effective way to introduce innovations to farmers and attract domestic and international financing. POA will initiate the annual event and transfer its leadership to the National Horticulture Trade Association, ensuring sustainability and lasting impact. Activity 3.6 will lead to Increased Availability of Improved Inputs (FFPr 1.2.1), Improved Capacity of Key Groups in the Agriculture Sector (FFPr 1.4.4), Increased Leverage of Private-Sector Resources (FFPr 1.4.5; 2.4.5).

POA will collaborate with the National Horticulture Trade Association, Mauritanian Investment Promotion Agency, Ministry of Agriculture, and Ministry of the Economy to organize the inaugural Annual Mauritanian Ag Expo in Q1 of Year 2. The event will bring farmers, investors, and agribusinesses together for a two-day gathering in Rosso. Women and youth farmers will be encouraged to participate in the event. This cost-effective, high-impact activity will expose farmers to innovations and technologies to enhance their output and sustainability, share lessons learned, and build connections (Figure 11). Domestic and international investors will also gain an understanding of the farmers' capacity and the overall market potential of the horticulture sector. In Q1 of Year 3, POA will transfer responsibility for the second Annual Agriculture Expo to the National Horticulture Trade Association. The National Horticulture Trade Association will generate income from the event by charging fees to presenters and exhibitors. The government will partner with the National Horticulture Trade Association and support the event with ministry resources.

Fig. 11: US Supplier Marketing Opps

Ag Expos are an excellent way to market innovative technologies to farmers to increase production and resilience. US suppliers are well-positioned to supply inputs, equipment, and machinery to Mauritania's emerging horticulture sector.

3.7: Build Government Capacity to Support Agriculture and Sustain Investments

- **Activity Lead:** POA
- **Target Regions:** Trarza, Brakna, Gorgol, and Nouakchott
- **Target Beneficiaries:** Agro-dealers, Government of Mauritania, and Farmers

As the Strategic Analysis indicates, the Government of Mauritania can initiate critical interventions to unlock productivity in the horticulture sector. To this end, POA proposes a cross-cutting approach to building capacity through GIS/remote sensing technology and

advocating for improved government policies. POA will collaborate with the Ministry of Agriculture to modify the existing input distribution and subsidization policy. This will stimulate private sector competition and enhance local capacity, quality, and timeliness of input availability. Activity 3.7 will lead to Improved Availability of Improved Inputs (FFPr 1.2.1), Increased Capacity of Government Institutions (FFPr 1.4.1; 2.4.1), and Improved Policy and Regulatory Framework (FFPr 1.4.2; 2.4.2).

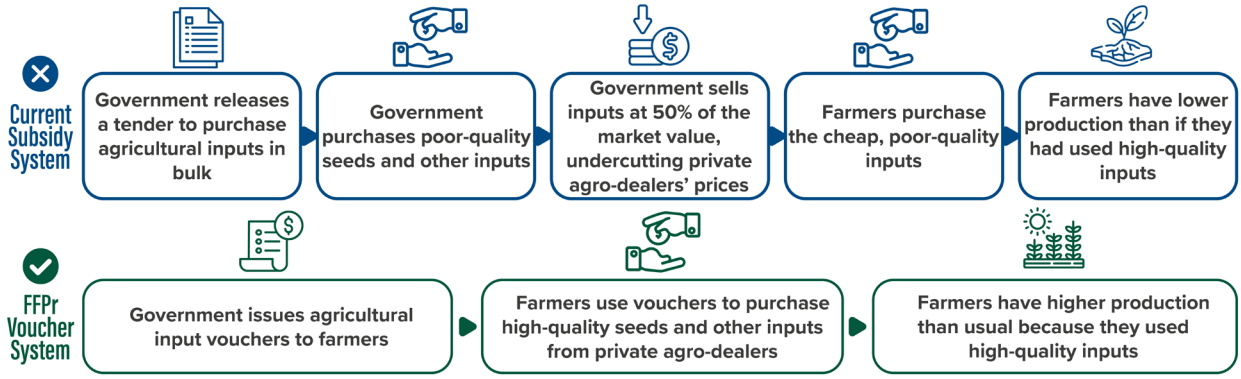
3.7.1: Develop the Government of Mauritania's Data Usage Capacity

In Q1 of Year 2, FFPr Mauritania will enhance the capacity of relevant government counterparts to use GIS and remote sensing data. POA will leverage data generated in Activity 1.2 and collaborate with the Ministry to develop plans to sustain GIS and remote sensing data. POA and 6th Grain will collaborate to boost the capacity of Ministry of Agriculture staff to use GIS and remote sensing data in Q2 of Years 3 and 4. Furthermore, POA will promote promising digital extension resources that the Ministry can use to support farmers. By integrating digital extension with GIS and remote sensing data, FFPr Mauritania aims to enhance the quality and quantity of available data to increase horticulture production in Mauritania.

3.7.2: Improve Subsidized Inputs Program

FFPr Mauritania will facilitate a discussion to modify the Ministry of Agriculture's input subsidy program by Q1 of Year 2. Based on discussions with Ministry officials and key stakeholders, including the CEO of Sadarane Farms, who is in continuous contact with the Minister of Agriculture, POA has assessed that the conditions are ripe for change. POA will collaborate with the Ministry of Agriculture and key stakeholders, such as the World Bank, the International Fund for Agricultural Development, and the Islamic Development Bank, to design and implement a voucher-based program (Figure 12).

Fig 12: Modifying the Government's Subsidy Program Will Increase Competition and Production



IV. Activity 4 – Build Capacity for Post-Harvest

4.1: Facilitate Investment Roundtables

- **Activity Lead:** POA
- **Target Regions:** Nouakchott and Rosso
- **Beneficiaries:** Investors, Farmers, and the Government of Mauritania

As the Strategic Analysis states, Mauritania's horticulture sector faces growth challenges due to insufficient investment in post-harvest storage, transformation, and end-market development. While investors are eager to enter the horticulture post-harvest sector, they cannot

determine the best potential investments due to the lack of reliable data and knowledge. This knowledge gap has become a significant barrier to entry, preventing potential investors from making informed decisions. To address this issue, POA will organize Investment Roundtables to bring together potential partners and leverage the data generated in Activity 1.2. Activity 4.1 will lead to Improved Linkages between Buyers and Sellers (FFPr 2.2.2), Increased Access to Improved Market Information (FFPr 1.4.3; FFPr 2.4.3), Increased Capacity of Government Institutions (FFPr 1.4.1; 2.4.1), Improved Capacity of Key Groups in the Agriculture Sector (FFPr 1.4.4), and Increased Leverage of Private Sector Resources (1.4.5; 2.4.5).

Incentivizing Foreign Investment

FFPr Mauritania will leverage Mauritania's commitment to building an attractive environment for investment. Mauritania's government incentivizes foreign investment by offering low interest rates and eight years of tax exoneration for new businesses. By supplying reliable data and building capacity, FFPr will optimize existing incentives to invest in the horticulture sector.

In Q4 of Year 1, POA will collaborate with the Mauritanian Agriculture Chamber of Commerce, Ministry of the Economy, US Mauritanian Business Council, and Mauritanian Investment Promotion Agency to organize Investment Roundtables in Nouakchott. These events will leverage the production data obtained through GIS and remote sensing (Activity 1.2) and present it to relevant agribusinesses and investors from the US, Mauritania, and other interested countries. The focus will be on fostering partnerships between US and Mauritania private sector investors interested in horticulture processing.

POA will lead a sustainability plan to ensure the continuation of the investment roundtables beyond the FFPr Mauritania project. This will involve working with key private sector partners and relevant public entities, such as the Ministry of Agriculture, the Mauritanian Agriculture Chamber of Commerce, and the US Embassy, to transfer the responsibility and develop sustainable finance mechanisms for the events. By Q2 of Year 5, POA will fully transfer organizational responsibility to these entities. The aim is to create a sustainable platform for increasing engagement between investors and stakeholders in the horticulture sector, fostering knowledge exchange and investment opportunities, and ultimately driving sector growth.

4.2: Increase Post-harvest Processing Capacity

- **Activity Lead:** POA
- **Target Regions:** Nouakchott, Trarza, and Brakna
- **Beneficiaries:** Local and International Investors, Agribusinesses, and Farmers

As described in the Strategic Analysis, the absence of post-harvest processing in the Mauritanian horticulture sector is a significant challenge. Without processing opportunities, substantial quantities of horticulture production are lost to spoilage. To address this challenge, POA has identified two strategic areas to invest in post-harvest processing capacity: large-scale export-oriented facilities and domestic market operations. POA will release tenders for these two levels of processors and co-invest in crucial infrastructure to boost domestic food transformation. FFPr Mauritania will train these two categories of processors in relevant food safety certifications and guidelines, including HAACP, GAP, and GFSI. These food safety standards and certifications will be the foundation for Mauritania's future horticulture export market. Activity 4.2 will lead to Increased Adoption of Established Standards by Industry (FFPr 2.1.1.1), Increased Use of Improved Post-Production Processing and Handling Practices (FFPr 2.1.2.1), Improved Post-Harvest Infrastructure (FFPr 2.1.2.2), and Increased Leverage of Private Sector Resources (1.4.5; 2.4.5).

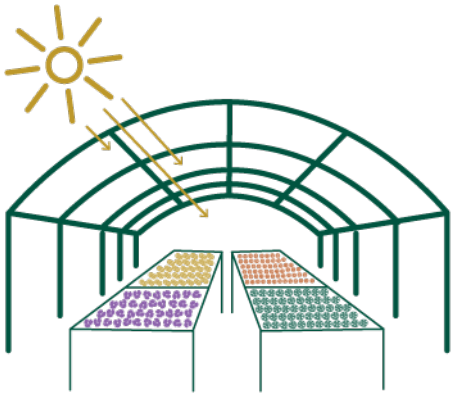
4.2.1: Incentivize Export-Oriented Processing

In Q3 of Year 2, POA will develop and release tenders to co-invest in export-scale processing infrastructure. FFPr Mauritania will evaluate proposals from local and foreign investors who are expected to invest at a 3:1 or 4:1 cost share (i.e., one dollar of FFPr Mauritania programs will match three to four dollars of private sector investment). To ensure that investors understand the high volumes of production to expect, the tender will include production increases and models of key crops resulting from the program. Co-investors will be encouraged to propose centrally located sites for processing facilities, such as Nouakchott and Rosso, to ensure easy access to roads, ports, and services to optimize trade. A bridge from Rosso south to Senegal is expected to be completed in 2024, providing increased opportunities for regional trade of processed goods. Target products may include tomato sauce, sun-dried tomatoes, and fruit juices. These opportunities are already under discussion, and the co-investors will be expected to bring forward similar proposals for investment through this opportunity.

4.2.2: Build Domestic-Oriented and On-Farm Processing Capacity

Women's cooperatives will have newfound opportunities to increase their processing activities due to larger production volumes. Currently, many of these groups are involved in small-scale processing activities. Targeted investment and support will help to add value to their existing products. With access to innovative processing equipment and technology, these groups will expand their operations and increase their processing capacity and quality. In Q4 of Year 2, POA will collaborate with promising women's cooperatives and large-scale farmers to co-invest in small-scale processing infrastructure. This infrastructure will enable the processing of locally sourced products for small villages and regional Valley hubs like Rosso, Bogue, Kaedi, Kiffa, and Nouakchott. To support these processing activities, POA will provide proven post-harvest processing tools, such as the United Nations QSM and Associates solar dryer, designed for cottage industry products (Figure 13). These processing facilities will improve farmers' and cooperatives' livelihoods by creating shelf-stable products with viable end-markets.

Fig 13: Green Energy Processing



POA will work with women's cooperatives to use cost-effective processing methods, such as solar dryers and food processors.

4.2.3: Strengthen Food Safety Capacity for Post-Harvest Processing and Transport

Starting in Q4 of Year 2, POA will launch an extensive food safety training program for farmers and processors. Capacity building will cover all aspects of the food supply chain, including harvest, packaging, transport, and storage. The program will focus on critical food safety practices and certifications such as HAACP, GFSI, and GAP, essential for domestic and export markets. POA will collaborate with the Ministry of Agriculture to enhance its institutional knowledge of food safety to ensure the effectiveness and sustainability of the training program. This partnership will enable the Ministry of Agriculture to support processors after FFPr Mauritania ends.

4.3: Increase Green Energy-Powered Cold and Dry Storage Capacity

- **Activity Lead:** POA

- **Target Regions**: Trarza (Rosso), Brakna (Bogue), Gorgol (Kaedi), and Nouakchott
- **Beneficiaries:** Farmers, Agribusiness, and Domestic Investors

As stated in the Strategic Analysis, the growth of the Mauritanian horticulture sector hinges on increasing cold and dry storage capacity. Domestic and foreign investors are interested in setting up cold storage facilities for crops grown in Mauritania and imported goods from Morocco, Europe, and the US. However, it is essential to conduct market research and analysis to ensure the feasibility and profitability of such an investment. To overcome this challenge, POA plans to collaborate with the Global Cold Chain Alliance and conduct a comprehensive Country-level Green Energy Cold Chain Feasibility Study. POA will leverage this study to incentivize investment in 13.8K cubic meters of dry storage and 9.2K cubic meters of cold storage with interested agribusinesses. Activity 4.3 will lead to Increased Use of Improved Post-Production Processing and Handling Practices (FFPr 2.1.2.1), Improved Post-Harvest Infrastructure (FFPr 2.1.2.2), and Increased Leverage of Private Sector Resources (1.4.5. 2.4.5).

Aldelano Solar Systems

POA is connected to Aldelano, a Black-owned business based in Jackson, Tennessee. Aldelano constructs 20-40 ft solar cold boxes that have the capacity to cold store up to 22,000 kg of food. The Aldelano ColdBox made in Tennessee is a promising on-farm storage facility for Mauritanian farmers.

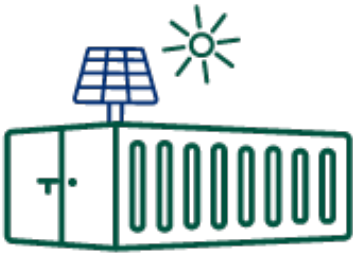
4.3.1: Conduct Country-Level Cold Chain Feasibility Study

In Q4 of Year 1, POA and the Global Cold Chain Alliance will collaborate on a Country-level Cold Chain Feasibility Study in Mauritania. The study will leverage each organization's expertise to identify and validate opportunities for private sector investment in cold chain facilities. Specifically, the Global Cold Chain Alliance will partner with POA and contribute design expertise to draft preliminary facilities, identify ideal locations, and ensure the feasibility of the identified opportunities. The study will emphasize climate-controlled cold and dry storage for FFPr Mauritania target crops and products such as watermelon and dates. The study will also analyze two-way trade prospects to facilitate a steady food supply to the market. This study will provide valuable insights into the market conditions and growth opportunities in Mauritania's cold chain industry, which will help attract investors.

4.3.2: Incentivize Large-Scale Cold and Dry Storage

Using information from the Cold Chain Feasibility Study, POA will release a solicitation for co-investors in cold and dry storage in Q1 of Year 2. FFPr Mauritania will solicit proposals for storage in important markets (Moroccan Mosque Market in Nouakchott), ports, and critical regional hubs such as Rosso and Bogue. POA has identified promising Mauritanian private sector partners interested in co-investing in large-scale cold and dry storage facilities, including Africom SARL, WAFA Holding, Rim Fruit, and the National Horticulture Trade Association.

Fig 14: On-Farm Storage



Women's cooperatives will be empowered through solar-powered storage units. These shipping-container-sized units will extend the shelf-life for their and nearby farmers' yields.

4.3.3: Grow Small-Scale Cold Storage Capacity

In Q1 of Year 1, POA will conduct a baseline assessment and select roughly ten women's cooperatives to cost-share on-farm cold and dry storage units (Figure 14). Partnering with the Global Cold Chain Alliance and its network of innovative companies, POA will identify cold and dry storage units best suited for Mauritania's small-scale farmers. POA has identified one high-

potential technology: the Aldelano Solar ColdBox Storage, an innovative, green energy-powered, farm-level storage solution. In Year 2, POA will train the cooperatives operating and maintaining the storage units. With increased cold and dry storage capacity, these cooperatives can expand their crop sales period and maintain processing operations throughout the year.

4.4: Facilitate Trade Show Attendance

- **Activity Lead:** POA
- **Target Regions:** International
- **Beneficiaries:** Processors and Commercial Farmers

As the Strategic Analysis mentions, Mauritania has the potential to market fresh and processed horticulture products globally. For this to happen, investors must gain exposure to international standards and buyers. POA will facilitate groups to attend global food trade shows. Activity 4.4 will lead to Improved Linkages between Buyers and Sellers (FFPr 2.2.2), Improved Marketing of Agricultural Products (FFPr 2.1.3; 2.2.1), and Increased Access to Improved Market Information (FFPr 1.4.3; FFPr 2.4.3).

Starting in Q1 of Year 3, POA will send groups, including agribusiness hubs, large-scale and domestic processors, and commercial farmers, to global food trade shows. Trade shows will include SIAL Paris and Summer Fancy Food in the US. The goal of participating in the events will be to connect, negotiate, and establish partnerships with buyers and suppliers.

4.5: Reduce Waste to Improve Regenerative Agriculture

- **Activity Lead:** POA
- **Target Regions:** Trarza, Brakna, Gorgol, Assaba, and Guidimakha
- **Target Beneficiaries:** Farmers, Investors, and Agribusinesses

As stated in the Strategic Analysis, Mauritania is currently unprepared to handle the significant organic waste that export-oriented food processors will generate. To address this, FFPr Mauritania will reduce horticulture sector waste by incentivizing businesses that convert waste into profitable products. These new businesses will enable scaling regenerative agriculture practices throughout the country. Activity 4.5 will lead to Increased Availability of Improved Inputs (FFPr 1.2.1) and Increased Leverage of Private Sector Resources (1.4.5).

In Q3 of Year 2, FFPr Mauritania will issue calls for proposals from entrepreneurs to start innovative waste transformation businesses. POA will incentivize businesses that convert agricultural waste into biofuels, organic fertilizers, and compost products. These businesses will use waste from new processing activities and by-products from crop production (Figure 15). For example, new processing companies such as sun-dried tomato canning factories will produce high volumes of waste. Investors can also take advantage of the waste already produced by the agricultural industry, like the leftover materials from large-scale rice production. FFPr Mauritania will sensitize farmers on the benefits of products such as organic fertilizer through Activity 1.1 to ensure a viable domestic end-market for new products.

Fig 15: Waste Transformation Enterprises

POA will incentivize the creation of businesses that transform organic waste into high-quality products. Businesses collect organic waste, such as spoiled food, from markets and byproducts at processing facilities. They bring them to a central location and use innovative transformation techniques to create products. The products are then sold to agro-dealers and farmers to aid regenerative agriculture production.